

tf.fingerprint Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/fingerprint

Generates fingerprint values.

Function Signatures

```
tf.fingerprint( data, method='farmhash64', name=None )
```

Code Examples

```
tf.fingerprint(  
data, method='farmhash64', name=None  
)
```

```
tf.fingerprint(  
data, method='farmhash64', name=None  
)
```

```
tf.fingerprint(data) == tf.fingerprint(tf.reshape(data, ...))  
tf.fingerprint(data) == tf.fingerprint(tf.bitcast(data, ...))
```

```
tf.fingerprint(data) == tf.fingerprint(tf.reshape(data, ...))  
tf.fingerprint(data) == tf.fingerprint(tf.bitcast(data, ...))
```

Documentation

Generates fingerprint values.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.fingerprint`

Generates fingerprint values of data.

Fingerprint op considers the first dimension of data as the batch dimension, and `output[i]` contains the fingerprint value generated from contents in `data[i, ...]` for all `i`.

Fingerprint op writes fingerprint values as byte arrays. For example, the default method `farmhash64` generates a 64-bit fingerprint value at a time. This 8-byte value is written out as an `tf.uint8` array of size 8, in little-endian order.

For example, suppose that data has data type `tf.int32` and shape `(2, 3, 4)`, and that the fingerprint method is `farmhash64`. In this case, the output shape is `(2, 8)`, where 2 is the batch dimension size of data, and 8 is the size of each fingerprint value in bytes. `output[0, :]` is generated from 12 integers in `data[0, :, :]` and similarly `output[1, :]` is generated from other 12 integers in `data[1, :, :]`.

Note that this op fingerprints the raw underlying buffer, and it does not fingerprint Tensor's metadata such as data type and/or shape. For example, the fingerprint values are invariant under reshapes and bitcasts as long as the batch dimension remain the same:

For string data, one should expect `tf.fingerprint(data) != tf.fingerprint(tf.string.reduce_join(data))` in general.

Returns